

IDENTIFICATION

Product Code: MAINDEC 08-D02B-D
Product Name: PDP-8 Instruction Test Part 2B
Date Created: January 12, 1968
Maintainer: Diagnostic Group

1. ABSTRACT

This program is a test of the 2s complement add (TAD) and rotate logic (RAL, RTL, RAR, RTR). Random numbers are used in the Twos Add portion of the test and sequential numbers are used in the Rotate portion. Program control depends on operator manipulation of four switches in the SWITCH REGISTER (bits 0, 1, 2, 3). Error information is normally printed out on the keyboard printer.

2. REQUIREMENTS

Storage

Memory locations 20_8 - 4177_8 .

Subprograms and/or Subroutines

High RIM Loader, High Binary Loader.

Equipment

PDP-8 Processor-Keyboard Reader

3. USAGE

3.1 Loading

If the Binary Loader beginning at 7777_8 is in memory, load the Instruction Test - Part 2b. Otherwise, the RIM Loader beginning at 7756_8 and/or the Binary Loader must be loaded into memory.

PDP-8 Instruction Test - Part 2B (Maindec 801-2B) may now be loaded as follows:

Set 7777_8 in the SWITCH REGISTER.

Press LOAD ADDRESS key.

Place Instruction Test-Part 2B in the keyboard reader.

Press START key on the operator console.

Engage the keyboard reader.

3.2 Switch Settings

When starting at the TAD portion (200_8) of the test, set switches 0 and 2 to the 1 state. This switch configuration allows the program to print any error message and halt on the error condition. After the TAD portion has run for a minimum of 10 minutes, set switch 3 to a 1 to enter the Rotate Test.

When starting at the rotate portion (2000_8) set switches 0 and 2 to the 1 state as above. This switch configuration allows the program to print any error message and halt on the error condition.

Switch 0	Stop on error (406_8 for TAD or 2433_8 for Rotate Test).
Switch 1	Scope mode (repeat loop causing the error).
Switch 2	Print error.
Switch 3	Leave the Twos Add test and start the Rotate Test.

Switch 0 and 1 Scope mode and stop on error.
 Switch 0 and 2 Print error and halt.
 Switch 1 and 2 Scope and print error.

3.3 Start-Up and/or Entry

The starting address of the TAD portion of the test is 0200_g. The starting address of the Rotate portion of the test is 2000_g. If bit 3 of the SWITCH REGISTER is set, it automatically causes an exit from the Twos Add portion of the test to the Rotate portion of the test.

Set either 0200_g in the SWITCH REGISTER to start at the Twos Add portion of the test, or set 2000_g in the SWITCH REGISTER to start at the Rotate portion of the test.

Press the LOAD ADDRESS key.

Press the START key.

3.4 Errors in Usage

The error halt for TAD Test is 406_g.

The error halt for Rotate Test is 2433_g.

Error printouts from both tests would appear as follows:

TWOS ADD ERROR PRINTOUT:

Good	Bad	X ARG	Y ARG
0 000000000001	0 000000000000	0 000000000000	000000000001

Indicating loss of a 1 bit in AC bit 11.

ROTATE ERROR PRINTOUTS:

PAT 0	000000000001	(original pattern)
RAL 0	000000000010	(pattern after RAL inst.)
RAR 0	000000000000	(pattern after RAR inst.)

Indicating loss of a 1 bit in AC bit 11 as a result of an RAR.

PAT 0	000000100000
RTR 0	000000000000
RTL 0	000000000000

Indicating loss of a 1 bit in AC bit 8 as a result of an RTR.

3.5 Recovery from such Errors

The program may be continued after it halts on an error, by pressing the CONTINUE key. The program continues to the next test, unless scope mode (bit 1) is requested.

Set the state of AC switch 1 to 1 to repeat the loop causing the error (scope mode).

Reference 4.3 for other switch variations.

4. RESTRICTIONS

This test should be run only after a successful run of the Instruction Test 2A to provide maximum reliability of the module repair table.

5. DESCRIPTION

5.1 Discussion

The PDP-8 Instruction Test-Part 2B tests the 2s ADD and ROTATE logic.

The 2s ADD logic is tested by the addition of pseudo random numbers. Two pseudo random numbers are generated and 2s added by a logical (simulated) adder. The same two numbers are added by the 2s add logic (TAD). The results are compared, and if an equality exists, two new random numbers are generated and the sequence is re-executed. If an inequality exists, the computer halts and/or types the error condition depending on the switch settings.

5.2 Examples and/or Applications

The error printout will contain the correct answer, the incorrect answer, and the two random numbers used.

Visual inspection of these patterns will determine the cause of the error. A lookup table is provided for rapid repair which will give all of the information shown in section 4.6.

Exit from TAD Test to the Rotate portion is accomplished by setting bit 3 in the SWITCH REGISTER. This switch also causes the program to print "ADD OK."

The Rotate Test generates 8192 patterns to be tested on two pairs of rotate instructions. The first pair of rotate instructions to be tested is RAL and RAR. The test pattern is rotated left once, then the result is rotated right once. The following items are compared:

The result of the RAR should equal the test pattern and original link.

The result of the link after the RAL should equal bit 0 of the test pattern.

If the RAR results and link equals the test pattern and link, the RAL and RAR instructions have operated correctly.

If an error occurs and an error printout is requested, the test pattern and the results of both the RAL and RAR instructions are printed. Visual inspection of these patterns will determine the probable cause of the error.

The second pair of rotate instructions to be tested is RTR and RTL. The test pattern is rotated right twice, then the result is rotated left twice. The following items are compared:

The result of the RTL should equal the test pattern and original link.

The result of the link after the RTR should equal pattern bit 1 of the test pattern.

If the RTL results and link equal the test pattern and link, the RTR and RTL, instructions have operated correctly.

If an error occurs and an error printout is requested, the test pattern and the results of both the RTR and RTL instructions are printed. Visual inspection of these patterns will determine the probable cause of the error.

After a complete pass through the Rotate Test, the computer will print ROT.

A printout of "2B" indicates the completion of a complete pass through the entire set of tests, after which the test begins again.

6. METHODS

See description section 5.

7. EXECUTION TIME

The TAD section takes 1 second for one complete pass; it will cycle continuously unless AC switch 3 is set. The Rotate portion takes 3 seconds for one complete pass.

8. PROGRAM LISTING

/PDP-8 INSTRUCTION TEST PART 2B ADD-ROTATE

0000 0000
0000 0000
0001 5001
0002 0002
0003 0003

*0
0000
JMP 1
2
3

0020 0020
0020 0000
0021 6046
0022 6041
0023 5022
0024 7200
0025 5420

*0020
PRXLOP, 0 /PRINT LOOP
TLS
LPXX, TSF
JMP LPXX
CLA
JMP I PRXLOP

0026 0000
0027 7240
0030 0104
0031 4020
0032 7240
0033 0103
0034 4020
0035 7240
0036 0103
0037 4020
0040 5426

CRLFLF, 0
CLA CMA
AND CR /CR
JMS PRXLOP
CLA CMA
AND LF /LF
JMS PRXLOP
CLA CMA
AND LF /LF
JMS PRXLOP
JMP I CRLFLF

0041 0000
0042 7240
0043 0104
0044 4020
0045 7240
0046 0103
0047 4020
0050 5441

CRLF, 0
CLA CMA
AND CR /CR
JMS PRXLOP
CLA CMA
AND LF /LF
JMS PRXLOP
JMP I CRLF

0051 0000
0052 0000
0053 0000
0054 0000
0055 0000
0056 0000

PAT, 0 /GENERATOR PATTERN
RALRTL, 0 /ROTATE LEFT PATTERNS
LFTLNK, 0 /ROTATE LEFT LINK PATTERNS
RARTRH, 0 /ROTATE RIGHT PATTERNS
RITLNK, 0 /ROTATE RIGHT LINK PATTERNS
TST1, 0 /TEST FLAG

0057	0000	PROUT,	0	/PRINT OUT LOCATION
0060	4000	K4000,	4000	/MASK LIST
0061	2000	K2000,	2000	
0062	1000	K1000,	1000	
0063	0400	K0400,	0400	
0064	0200	K0200,	0200	
0065	0100	K0100,	0100	
0066	0040	K0040,	0040	
0067	0020	K0020,	0020	
0070	0010	K0010,	0010	
0071	0004	K0004,	0004	
0072	0002	K0002,	0002	
0073	0001	K0001,	0001	
0074	0057	XPROUT,	PROUT	
0075	0322	R,	0322	/R
0076	0301	A,	0301	/A
0077	0314	L,	0314	/L
0100	0324	T,	0324	/T
0101	0320	P,	0320	/P
0102	0240	SP,	0240	/SP
0103	0212	LF,	0212	/LF
0104	0215	CR,	0215	/CR
0105	0060	ZERO,	0060	/ZERO
0106	0061	ONE,	0061	/ONE
0107	0317	O,	0317	/O ALPHA
0110	0313	K,	0313	/K
0111	7764	COUNT,	7764	/MINUS 11
0112	0000	STRCNT,	0	
0113	0262	TWO,	0262	/2
0114	0302	B,	0302	/B
0115	0000	WD1,	0	
0116	0000	WD2,	0	
0117	0000	BW1,	0	
0120	0000	CRY,	0	
0121	0000	TOTAL,	0	
0122	0000	SUM,	0	
0123	0000	CNTR,	0	
0124	0000	HEADER,	0	
0125	0000	BITSTR,	0	
0126	7776	SPAC06,	7776	/MINUS 1
0127	0000	SPACST,	0	
0130	0307	G,	0307	/G
0131	0304	D,	0304	/D
0132	0330	X,	0330	/X
0133	0331	Y,	0331	/Y
0134	0000	LINK,	0	/LINK
0135	0000	XARG,	0	/XARG
0136	0000	YARG,	0	/YARG
0137	7763	COUNTX,	7763	
0140	0000	LNKSTR,	0	
0141	7377	K7377,	7377	

39<2	0000	0		
0143	7240		CLA CMA	
0144	0140		AND Z LNKSTR	
0145	7440		SZA	
0146	5150		JMP SL	
0147	5152		JMP CL	
0150	7360	SL,		CLA CMA STL
0151	5542		JMP I CX	
0152	7340	CL,		CLL CLA CMA
0153	5542		JMP I CX	

*4000

4000	7200	RAND2,	CLA	
4001	1417	TAD I 0017		
4002	3135	DCA XARG		/STORE FIXED PAT
4003	1417	TAD I 0017		
4004	3136	DCA YARG		/STORE FIXED PAT
4005	2216	ISZ RCNT		
4006	5647	JMP I XSTRXY		/EXIT TO TEST
4007	1215	TAD LISTX		
4010	3017	DCA 0017		
4011	1214	TAD M144		
4012	3216	DCA RCNT		
4013	5647	JMP I XSTRXY		/EXIT TO TEST
4014	7634	M144,	-144	
4015	4177	LISTX,	LIST-1	
4016	0000	RCNT,	0000	
4017	0000	ODEVEN,	0000	
4020	7300	RAND,	CLL CLA	/FIXED PATTERN
4021	2217	ISZ ODEVEN		/RANDOM PATTERN
4022	7000	NOP		
4023	1217	TAD ODEVEN		
4024	7010	RAR		
4025	7630	SZL CLA		
4026	5230	JMP RAND1		
4027	5200	JMP RAND2		
4030	7604	RAND1,	CLA OSR	
4031	0063	AND Z K0400		
4032	7000	NOP		
4033	7440	SZA		
4034	5650	JMP I ADDX		/SW 3 EQUALS A ONE TO EXIT
4035	7240	CLA CMA		
4036	0121	AND Z TOTAL		
4037	7000	NOP		
4040	3135	DCA Z XARG		
4041	7040	CMA		
4042	0121	AND Z TOTAL		
4043	7001	IAC		
4044	1410	TAD I Z 10		
4045	3136	DCA Z YARG		
4046	5647	JMP I XSTRXY		
4047	0225	XSTRXY,	STRXY	
4050	0312	ADDX,	PADDOK	

*0017
0017 4177 LIST-1

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*4051
4051 7240 FCOMP,      CLA CMA      /COMPARE SUM AND TOTAL
4052 0121   AND Z TOTAL
4053 7040   CMA
4054 0122   AND Z SUM
4055 3275   DCA CXM
4056 7240   CLA CMA
4057 0122   AND Z SUM
4060 7040   CMA
4061 0121   AND Z TOTAL
4062 3274   DCA CXN
4063 7240   CLA CMA
4064 0275   AND CXM
4065 7440   SZA
4066 5676   JMP I ERX              /ERROR
4067 7240   CLA CMA
4070 0274   AND CXN
4071 7440   SZA
4072 5676   JMP I ERX              /ERROR
4073 5277   JMP LCOMP
4074 0000   CXN,      0
4075 0000   CXM,      0
4076 0400   ERX,      ERROR
4077 7240   LCOMP,    CLA CMA      /COMPARE CRY AND LINK
4100 0134   AND Z LINK              /LINK BIT IN BIT 11
4101 7040   CMA
4102 0120   AND Z CRY
4103 3322   DCA LRX
4104 7240   CLA CMA
4105 0120   AND Z CRY
4106 7040   CMA
4107 0134   AND Z LINK
4110 3323   DCA LRY
4111 7240   CLA CMA
4112 0322   AND LRX
4113 7440   SZA
4114 5676   JMP I ERX              /ERROR
4115 7240   CLA CMA
4116 0323   AND LRY
4117 7440   SZA
4120 5676   JMP I ERX              /ERROR
4121 5724   JMP I NOERX
4122 0000   LRX,      0
4123 0000   LRY,      0
4124 0407   NOERX,    NOERR

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*4200

4200	7777	LIST,	7777	4262	7777	7777
4201	7777	7777		4263	0001	0001
4202	7776	7776		4264	7777	7777
4203	7777	7777		4265	0002	0002
4204	7775	7775		4266	7777	7777
4205	7777	7777		4267	0004	0004
4206	7773	7773		4270	7777	7777
4207	7777	7777		4271	0010	0010
4210	7767	7767		4272	7777	7777
4211	7777	7777		4273	0020	0020
4212	7757	7757		4274	7777	7777
4213	7777	7777		4275	0040	0040
4214	7737	7737		4276	7777	7777
4215	7777	7777		4277	0100	0100
4216	7677	7677		4300	7777	7777
4217	7777	7777		4301	0200	0200
4220	7577	7577		4302	7777	7777
4221	7777	7777		4303	0400	0400
4222	7377	7377		4304	7777	7777
4223	7777	7777		4305	1000	1000
4224	6777	6777		4306	7777	7777
4225	7777	7777		4307	2000	2000
4226	5777	5777		4310	7777	7777
4227	7777	7777		4311	4000	4000
4230	3777	3777		4312	0001	0001
4231	7777	7777		4313	7777	7777
4232	7777	7777		4314	0002	0002
4233	7777	7777		4315	7777	7777
4234	7776	7776		4316	0004	0004
4235	7777	7777		4317	7777	7777
4236	7775	7775		4320	0010	0010
4237	7777	7777		4321	7777	7777
4240	7773	7773		4322	0200	0200
4241	7777	7777		4323	7777	7777
4242	7767	7767		4324	0400	0400
4243	7777	7777		4325	7777	7777
4244	7757	7757		4326	0100	0100
4245	7777	7777		4327	7777	7777
4246	7737	7737		4330	0200	0200
4247	7777	7777		4331	7777	7777
4250	7677	7677		4332	0400	0400
4251	7777	7777		4333	7777	7777
4252	7577	7577		4334	1000	1000
4253	7777	7777		4335	7777	7777
4254	7377	7377		4336	2000	2000
4255	6777	6777		4337	7777	7777
4256	7777	7777		4340	4000	4000
4257	5777	5777		4341	7777	7777
4260	7777	7777				
4261	3777	3777				


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0235 0000 ADDISM,      0      /FAKE ADD
0236 7300 CLA CLL
0237 3121 DCA Z TOTAL
0240 3120 DCA Z CRY
0241 7040 CMA
0242 0111 AND Z COUNT      /MINUS 11
0243 3123 DCA Z CNTR

0244 7040 AISM,      CMA
0245 0115 AND Z WD1
0246 7010 RAR
0247 3115 DCA Z WD1
0250 7004 RAL
0251 3117 DCA Z BW1
0252 7040 CMA
0253 0116 AND Z WD2
0254 7010 RAR
0255 3116 DCA Z WD2
0256 7040 CMA
0257 0117 AND BW1

0260 7420 SNL
0261 5302 JMP DISM
0262 7450 SNA
0263 5305 JMP CISM
0264 7300 CLL CLA
0265 7040 AXISM,      CMA
0266 0120 AND Z CRY
0267 7010 RAR
0270 7040 CMA
0271 0117 AND Z BW1
0272 3120 BISM,      DCA Z CRY
0273 7040 CMA
0274 0121 AND Z TOTAL
0275 7010 RAR
0276 3121 DCA Z TOTAL
0277 2123 ISZ Z CNTR
0300 5244 JMP AISM
0301 5635 JMP I ADDISM
0302 7450 DISM,      SNA
0303 5265 JMP AXISM
0304 7220 CML CLA
0305 7040 CISM,      CMA
0306 0120 AND Z CRY
0307 7440 SZA
0310 7100 CLL
0311 5272 JMP BISM

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0312	4041	PADDOK,	JMS Z CRLF	/CR LF
0313	7240	CLA CMA		
0314	0076	AND Z A	/A	
0315	4020	JMS Z PRXLOP		
0316	7240	CLA CMA		
0317	0131	AND Z D	/D	
0320	4020	JMS Z PRXLOP		
0321	7240	CLA CMA		
0322	0131	AND Z D	/D	
0323	4020	JMS Z PRXLOP		
0324	7240	CLA CMA		
0325	0102	AND Z SP	/SP	
0326	4020	JMS Z PRXLOP		
0327	7240	CLA CMA		
0330	0107	AND Z O	/O	
0331	4020	JMS Z PRXLOP		
0332	7240	CLA CMA		
0333	0110	AND Z K	/K	
0334	4020	JMS Z PRXLOP		
0335	5736	JMP I ROTATE	/EXIT ADD TEST	
0336	2000	ROTATE,	GEN1	
0337	4051	XFCOMP,	FCOMP	

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*0400
0400 7604 ERROR,          CLA OSR          /READ IN SR
0401 7106   CLL RTL
0402 7510   SPA          /SW2 EQUALS A ONE TO PRINT
0403 4216   JMS PRINT    /JMS TO PRINT ROUTINE
0404 7604   CLA OSR
0405 7510   SPA          /SW0 EQUALS A ONE TO HALT
0406 7402   HLT          /HALT ON ERROR
0407 7604 NOERR,        CLA OSR
0410 7104   CLL RAL
0411 7510   SPA          /SW1 EQUALS A ONE TO SCOPE MODE
0412 5614   JMP I SXY     /SCOPE MODE
0413 5615   JMP I INCRT   /CONTINUE MODE
0414 0225   SXY,          STRXY
0415 0223   INCRT,        INCR

0416 0000   PRINT,       0
0417 7240   CLA CMA
0420 0124   AND Z HEADER /HEADER FLAG
0421 7440   SZA
0422 4321   JMS PRHEAD   /JMS TO PRINT HEADER ROUTINE
0423 7000   PRERR,       NOP
0424 4041   JMS Z CRLF   /CR LF
0425 4020   JMS Z PRXLOP
0426 7240   CLA CMA
0427 0120   AND Z CRY
0430 4635   JMS I XONZER /TEST FAKE LINK FOR SEX AND
                                /PRINT A ONE OR ZERO

0431 7240   CLA CMA
0432 0102   AND Z SP     /PRINT SP
0433 4020   JMS Z PRXLOP
0434 5236   JMP PTOTAL   /PRINT CONTENTS OF FAKE ADD
0435 2637   XONZER,      ONZER

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0436	7240	PTOTAL,	CLA CMA	
0437	0121	AND Z TOTAL		/STORE CONTENTS OF FAKE ADD
0440	3125	DCA Z BITSTR		
0441	4266	JMS MESSG		
0442	7240	CLA CMA		
0443	0134	AND Z LINK		/TEST REAL LINK FOR SEX AND
0444	4635	JMS I XONZER		/PRINT A ONE OR ZERO
0445	7240	CLA CMA		
0446	0102	AND Z SP		/ PRINT SP
0447	4020	JMS Z PRXLOP		
0450	5251	JMP XTOTAL		
0451	7240	XTOTAL,	CLA CMA	
0452	0122	AND Z SUM		
0453	3125	DCA Z BITSTR		/STORE CONTENTS OF REAL ADD
0454	4266	JMS MESSG		
0455	7240	CLA CMA		
0456	0135	AND Z XARG		
0457	3125	DCA Z BITSTR		/STORE XARG
0460	4266	JMS MESSG		
0461	7240	CLA CMA		
0462	0136	AND Z YARG		
0463	3125	DCA Z BITSTR		/STORE Y ARG
0464	4266	JMS MESSG		
0465	5616	JMP I PRINT		/EXIT TO SWITCH ROUTINE

```

0466 0000 MESSG,      0
0467 7240   CLA CMA
0470 0137   AND Z COUNTX
0471 3112   DCA Z STRCNT
0472 2112   NBIT,      ISZ Z STRCNT
0473 7410   SKP
0474 5312   JMP PRSPAC      /12 COUNTS FINISHED
0475 7240   CLA CMA
0476 0125   AND Z BITSTR
0477 7100   CLL
0500 7004   RAL
0501 3125   DCA Z BITSTR      /STORE ROTATED WORD
0502 7430   SZL
0503 5306   JMP PRONE
0504 4764   PRZERO,      JMS I XZEROR      /PRINT ZERO
0505 5272   JMP NBIT
0506 7240   PRONE,      CLA CMA
0507 0106   AND Z ONE
0510 4020   JMS Z PRXLOP      /PRINT ONE
0511 5272   JMP NBIT
0512 7240   PRSPAC,      CLA CMA
0513 0102   AND Z SP
0514 4020   JMS Z PRXLOP      /SP
0515 7240   CLA CMA
0516 0102   AND Z SP      /SP
0517 4020   JMS Z PRXLOP
0520 5666   JMP I MESSG

0521 0000 PRHEAD,      0
0522 7200   CLA
0523 3124   DCA Z HEADER      /CLEAR HEADER FLAG
0524 7240   CLA CMA
0525 0126   AND Z SPAC06
0526 3127   DCA Z SPACST      /STORE SPACE COUNT
0527 4041   JMS Z CRLF      /PRINT CR LF

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0530 7240 SPA06,      CLA CMA
0531 0102      AND Z SP
0532 4020      JMS Z PRXLOP      /PRINT 6 SPACES
0533 2127      ISZ Z SPACST
0534 5330      JMP SPA06
0535 7240      CLA CMA
0536 0130      AND Z G          /G
0537 4020      JMS Z PRXLOP
0540 7240      CLA CMA
0541 0107      AND Z O          /O ALPHA
0542 4020      JMS Z PRXLOP
0543 7240      CLA CMA
0544 0107      AND Z O          /O ALPHA
0545 4020      JMS Z PRXLOP
0546 7240      CLA CMA
0547 0131      AND Z D          /D
0550 4020      JMS Z PRXLOP
0551 4762      JMS I MANYSP      /JMP TO PRINT 12 SPACES
0552 7240      CLA CMA
0553 0114      AND Z B          /B
0554 4020      JMS Z PRXLOP
0555 7240      CLA CMA
0556 0076      AND Z A          /A
0557 4020      JMS Z PRXLOP
0560 5761      JMP I CONHED
0561 0600      CONHED,      HEDCON
0562 0626      MANYSP,      TWELVE
0563 5721      HEDRJ,      JMP I PRHEAD      /EXIT HEADER ROUTINE
0564 2702      XZEROR,      ZEROR

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*0600
0600 7240 HEDCON,      CLA CMA
0601 0131 AND Z D      /D
0602 4020 JMS Z PRXLOP
0603 4226 JMS TWELVE   /12 SPACES
0604 7240 CLA CMA
0605 0132 AND Z X      /X
0606 4020 JMS Z PRXLOP
0607 7240 CLA CMA
0610 0102 AND Z SP     /SP
0611 4020 JMS Z PRXLOP
0612 4240 JMS ARGXXX   /ARG
0613 4226 JMS TWELVE   /12 SPACES
0614 7240 CLA CMA
0615 0133 AND Z Y      /Y
0616 4020 JMS Z PRXLOP
0617 7240 CLA CMA
0620 0102 AND Z SP     /SP
0621 4020 JMS Z PRXLOP
0622 4240 JMS ARGXXX   /ARG
0623 4041 JMS Z CRLF   /CR LF
0624 5625 JMP I RJHED  /JUMP TO EXIT HEADER ROUTINE
0625 0563 RJHED,      HEDRJ
0626 0000 TWELVE,     0
0627 7240 CLA CMA
0630 0111 AND Z COUNT
0631 3127 DCA Z SPACST /STORE MINUS 12
0632 7240 SPA12,      CLA CMA
0633 0102 AND Z SP     /SP
0634 4020 JMS Z PRXLOP /PRINT 12 SPACES
0635 2127 ISZ Z SPACST
0636 5232 JMP SPA12
0637 5626 JMP I TWELVE

0640 0000 ARGXXX,     0
0641 7240 CLA CMA
0642 0076 AND Z A      /A
0643 4020 JMS Z PRXLOP
0644 7240 CLA CMA
0645 0075 AND Z R      /R
0646 4020 JMS Z PRXLOP
0647 7240 CLA CMA
0650 0130 AND Z G      /G
0651 4020 JMS Z PRXLOP
0652 5640 JMP I ARGXXX

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*2000
2000 4316 GEN1,      JMS HSEKP
2001 4142 CONT1,    JMS Z CX
2002 0051      AND Z PAT
2003 7001      IAC
2004 3051      DCA Z PAT      /STORE INCREMENTED PATTERN
2005 7420      SNL
2006 5215      JMP CLRLNK      /JMP TO CLEAR LNKSTR
2007 1060      TAD K4000
2010 3140      DCA Z LNKSTR    /SET LNKSTR TO 4000
2011 4352 PT1EX,    JMS EX
2012 7440      SZA
2013 5220      JMP ROT1
2014 5274      JMP GEN2      /EXIT ROT1
2015 7200      CLRLNK,      CLA
2016 3140      DCA Z LNKSTR
2017 5211      JMP PT1EX

2020 7240      ROT1,      CLA CMA
2021 3056      DCA Z TST1      /SET TST1 FLAG
2022 7340      CLL CLA CMA
2023 0140      AND Z LNKSTR
2024 7440      SZA
2025 5272      JMP SETLNK
2026 7140      CLL CMA      /CLEAR LINK
2027 0051      REROT1,    AND Z PAT      /BRING UP PATTERN
2030 7004      RAL
2031 3052      DCA Z RALRTL    /STORE RAL PATTERN
2032 7430      SZL      /SKIP IF LINK EQUALS A ZERO
2033 1060      TAD Z K4000    /SET RAL LINK STORE
2034 3053      DCA Z LFTLNK    /CLEAR RAL LINK STORE
2035 7240      CLA CMA
2036 0052      AND Z RALRTL
2037 7010      RAR
2040 3054      DCA Z RARTR      /STORE RAR PATTERN
2041 7430      SZL      /SKIP IF LINK EQUALS A ZERO
2042 1060      TAD Z K4000    /SET RAR LINK STORE
2043 3055      DCA Z RITLNK    /CLEAR RAR LINK STORE

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2044	7340	CLL CLA CMA	
2045	0054	AND Z RARRTR	/RARRTR SHOULD EQUAL PAT
2046	7040	CMA	
2047	1051	TAD Z PAT	/COMPARE RARRTR WITH PAT
2050	7040	CMA	/AC SHOULD EQUAL ZERO
2051	7450	SNA	
2052	7430	SZL	
2053	5715	JMP I ERSWIX	/JUMP TO ERROR SWITCHES
2054	1060	TAD K4000	
2055	0051	AND Z PAT	/MASK BIT 0 OF PAT
2056	7040	CMA	
2057	1053	TAD Z LFTLNK	/COMPARE LFTLNK WITH PAT
2060	7040	CMA	/BIT 0
2061	7440	SZA	
2062	5715	JMP I ERSWIX	/JUMP TO ERROR SWITCHES
2063	1055	TAD Z RITLNK	
2064	7040	CMA	
2065	1140	TAD Z LNKSTR	/COMPARE PAT LINK WITH RITLNK
2066	7040	CMA	
2067	7440	SZA	
2070	5715	JMP I ERSWIX	
2071	5751	JMP I SXOKX1	
2072	7360	SETLNK, CLA CMA STL	/SET LINK
2073	5227	JMP REROT1	
2074	4316	GEN2, JMS HSEKP	
2075	4142	CONT2, JMS Z CX	
2076	0051	AND Z PAT	
2077	7001	IAC	
2100	3051	DCA Z PAT	/STORE INCREMENTED PATTERN
2101	7420	SNL	
2102	5311	JMP CLLINK	/JUMP TO CLEAR LNKSTR
2103	1060	TAD K4000	
2104	3140	DCA Z LNKSTR	/SET LNKSTR TO 4000
2105	4363	PT1EXX, JMS EX1	
2106	7440	SZA	
2107	5714	JMP I ROT2X	
2110	5332	JMP ROTOK	/EXIT ROTATE TESTS

2111	7200	CLLINK,	CLA	
2112	3140	DCA Z	LNKSTR	
2113	5305	JMP	PT1EXX	
2114	2200	ROT2X,	ROT2	
2115	2400	ERSWIX,	ERRSW1	
2116	0000	HSEKP,	0	
2117	7300	CLA	CLL	
2120	3051	DCA Z	PAT	
2121	3052	DCA Z	RALRTL	
2122	3054	DCA Z	RARRTR	
2123	3053	DCA Z	LFTLNK	
2124	3055	DCA Z	RITLNK	
2125	3140	DCA Z	LNKSTR	
2126	7000	NOP		
2127	7000	NOP		
2130	7000	NOP		
2131	5716	JMP I	HSEKP	
2132	7200	ROTK,	CLA	
2133	4041	JMS Z	CRLF	/CRLF
2134	1075	TAD Z	R	/R
2135	4020	JMS Z	PRXLOP	
2136	1107	TAD Z	0	/0
2137	4020	JMS Z	PRXLOP	
2140	1100	TAD Z	T	/T
2141	4020	JMS Z	PRXLOP	
2142	4041	JMS Z	CRLF	/CRLF
2143	1113	TAD Z	TWO	/2
2144	4020	JMS Z	PRXLOP	
2145	1114	TAD Z	B	/B
2146	4020	JMS Z	PRXLOP	
2147	5750	JMP I	ARITH	
2150	0200	ARITH,	ARITHT	
2151	2521	SXOKX1,	SWOKX1	
2152	0000	EX,	0	
2153	1140	TAD Z	LNKSTR	
2154	7440	SZA		
2155	7410	SKP		
2156	5220	JMP	ROT1	
2157	7240	CLA	CMA	
2160	0051	AND Z	PAT	
2161	7040	CMA		
2162	5752	JMP I	EX	
2163	0000	EX1,	0	
2164	1140	TAD Z	LNKSTR	
2165	7440	SZA		
2166	7410	SKP		
2167	5714	JMP I	ROT2X	
2170	7240	CLA	CMA	
2171	0051	AND Z	PAT	
2172	7040	CMA		
2173	5763	JMP I	EX1	

*2200

2200	7300	ROT2,	CLA CLL	
2201	3056	DCA Z TST1		/CLEAR TEST FLAG
2202	7340	CLL CLA CMA		
2203	0140	AND Z LNKSTR		
2204	7440	SZA		
2205	5250	JMP STLNK		
2206	7140	CLL CMA		
2207	0051	REROT2,	AND Z PAT	/BRING UP PATTERN
2210	7012	RTR		
2211	3054	DCA Z RARRTR		/STORE RTR PATTERN
2212	7430	SZL		/SKIP IF LINK EQUALS A ZERO
2213	1072	TAD Z K0002		/SET RTR LINK STORE
2214	3055	DCA Z RITLNL		/CLEAR RTR LINK STORE
2215	1054	TAD Z RARRTR		
2216	7006	RTL		
2217	3052	DCA Z RALRTL		/STORE RTL PATTERN
2220	7430	SZL		
2221	1060	TAD Z K4000		/SET RTL LINK STORE
2222	3053	DCA Z LFTLNL		/CLEAR RTL LINK STORE
2223	7100	CLL		
2224	1052	TAD Z RALRTL		/RALRTL SHOULD EQUAL PAT
2225	7040	CMA		
2226	1051	TAD Z PAT		/COMPARE RALRTL WITH PAT
2227	7040	CMA		
2230	7440	SZA		
2231	5652	JMP I ERSW2X		/JMP TO ERROR SWITCHES
2232	1072	TAD Z K0002		/COMPARE ROTLNL WITH PAT BIT 10
2233	0051	AND Z PAT		/MASK BIT 10 OF PAT
2234	7040	CMA		
2235	1055	TAD Z RITLNL		
2236	7040	CMA		
2237	7440	SZA		
2240	5652	JMP I ERSW2X		
2241	1053	TAD Z LFTLNL		/LFT LINK SHOULD EQUAL LNKSTR
2242	7040	CMA		
2243	1140	TAD Z LNKSTR		/COMPARE LFTLNL WITH LNKSTR
2244	7040	CMA		
2245	7440	SZA		
2246	5652	JMP I ERSW2X		/JUMP TO ERROR SWITCHES
2247	5653	JMP I SXOKX2		
2250	7360	STLNL,	CLA CMA STL	
2251	5207	JMP REROT2		
2252	2406	ERSW2X,	ERRSW2	
2253	2525	SXOKX2,	SWOKX2	

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*2400
2400 7200 ERRSW1,      CLA
2401 1244   TAD ROTX1
2402 3215   DCA ERIN      /SCOPE MODE RJMP ADDRESS
2403 1245   TAD CONTX1
2404 3214   DCA CONTX      /CONTINUE MODE RJMP ADDRESS
2405 5216   JMP ERSW
2406 7200 ERRSW2,      CLA
2407 1250   TAD ROTX2
2410 3215   DCA ERIN      /SCOPE MODE RJMP ADDRESS
2411 1251   TAD CONTX2
2412 3214   DCA CONTX      /CONTINUE MODE RJMP ADDRESS
2413 5216   JMP ERSW

2414 0000   CONTX,      0
2415 0000   ERIN,       0
2416 7604   ERSW,      CLA OSR      /READ IN SWITCHES
2417 0062   AND Z K1000      /MASK BIT 2
2420 7040   CMA
2421 1062   TAD Z K1000
2422 7040   CMA
2423 7450   SNA      /TEST BIT 2 SWITCH
2424 4255   JMS ROPR
2425 7604   CLA OSR
2426 0060   AND Z K4000      /MASK BIT 0
2427 7040   CMA
2430 1060   TAD Z K4000
2431 7040   CMA
2432 7450   SNA      /TEST BIT 0 SWITCH
2433 7402   HLT      /ERROR HALT
2434 7604   SWOK,      CLA OSR
2435 0061   AND Z K2000      /MASK BIT 1
2436 7040   CMA
2437 1061   TAD Z K2000
2440 7040   CMA
2441 7450   SNA      /TEST BIT 1 SWITCH
2442 5615   JMP I ERIN      /JMP TO SCOPE MOD
2443 5614   JMP I CONTX      /JMP TO CONTINUE MODE

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2444	2020	ROTX1,	ROT1	
2445	2001	CONTX1,	CONT1	
2446	2000	GEN1X1,	GEN1	
2447	2074	GEN2X2,	GEN2	
2450	2200	ROTX2,	ROT2	
2451	2075	CONTX2,	CONT2	
2452	2464	TWOROX,	TWOR0	
2453	2465	FINPRX,	FINPR	
2454	2650	RARPRX,	RARPR	
2455	0000	ROPR,	0	/RJMP TO SWITCH ROUTINE
2456	4026	JMS Z CRLFLF		/PRINT CR LF LF
2457	4714	JMS I PATPRX		/PRINT PAT
2460	7200	CLA		
2461	1056	TAD Z TST1		
2462	7440	SZA		
2463	5266	JMP ROT1PR		/PRINT ROTATE ONE PATTERN
2464	4715	TWOR0,	JMS I ROT2PX	/PRINT ROTATE TWO PATTERN
2465	5655	FINPR,	JMP I ROPR	
2466	7200	ROT1PR,	CLA	
2467	1254	TAD RARPRX		
2470	3714	DCA I PATPRX		
2471	4041	JMS Z CRLF		/PRINT CR LF
2472	7200	CLA		
2473	1075	TAD Z R		/R
2474	4020	JMS Z PRXLOP		
2475	1076	TAD Z A		/A
2476	4020	JMS Z PRXLOP		
2477	1077	TAD Z L		/L
2500	4020	JMS Z PRXLOP		
2501	1102	TAD Z SP		/SP
2502	4020	JMS Z PRXLOP		
2503	1053	TAD Z LFTLNK		
2504	7440	SZA		
2505	5716	JMP I LNONER		/LEFT LINK PRINT ONE
2506	4717	JMS I ZERORX		/LEFT LINK PRINT ZERO
2507	1102	RO1X,	TAD Z SP	
2510	4020	JMS Z PRXLOP		/SP
2511	1052	TAD Z RALRTL		
2512	3057	DCA Z PROUT		
2513	5720	JMP I COUNXX		/PRINT RALRTL CONTENTS
2514	2600	PATPRX,	PATPR	
2515	2732	ROT2PX,	ROT2PR	
2516	2676	LNONER,	LNONE	
2517	2702	ZERORX,	ZEROR	
2520	2616	COUNXX,	COUNPR	
2521	7200	SWOKX1,	CLA	
2522	1245	TAD CONTX1		
2523	3214	DCA CONTX		
2524	5234	JMP SWOK		
2525	7200	SWOKX2,	CLA	
2526	1251	TAD CONTX2		
2527	3214	DCA CONTX		
2530	5234	JMP SWOK		

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*2600
2600 0000 PATPR,      0
2601 1101 TAD Z P      /P
2602 4020 JMS Z PRXLOP
2603 1076 TAD Z A      /A
2604 4020 JMS Z PRXLOP
2605 1100 TAD Z T      /T
2606 4020 JMS Z PRXLOP
2607 1102 TAD Z SP     /SP
2610 4020 JMS Z PRXLOP
2611 4361 JMS PLINK
2612 1102 TAD Z SP
2613 4020 JMS Z PRXLOP /SP
2614 1051 TAD Z PAT
2615 3057 DCA Z PROUT  /STORE GENERATED PATTERN

2616 4231 COUNPR,      JMS MINDEX      /JMS TO MASK INDEX ROUTINE
2617 0137 AND Z COUNTX
2620 3112 DCA Z STRCNT
2621 2112 LSTBIT,      ISZ Z STRCNT
2622 7410 SKP
2623 5600 JMP I PATPR      /12 COUNTS FINISHED
2624 7200 CLA
2625 1057 TAD Z PROUT
2626 0410 AND I Z 10
2627 4237 JMS ONZER
2630 5221 JMP LSTBIT

2631 0000 MINDEX,      0
2632 7200 CLA
2633 1074 TAD Z XPROUT  /INDEX STARTING ADDRESS
2634 3010 DCA Z 10      /STORE INDEX ADDRESS
2635 7240 CLA CMA
2636 5631 JMP I MINDEX

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2637	0000	ONZER,	0	
2640	7440	SZA		
2641	5244	JMP ONEP		/JMP TO PRINT ONE
2642	4302	JMS ZEROR		
2643	5637	JMP I ONZER		
2644	7240	ONEP,	CLA CMA	
2645	0106	AND Z ONE		
2646	4020	JMS Z PRXLOP		/PRINT ONE
2647	5637	JMP I ONZER		
2650	7200	RARPR,	CLA	
2651	1273	TAD FINPRN		
2652	3200	DCA PATPR		
2653	4041	JMS Z CRLF		/CR LF
2654	7200	CLA		
2655	1075	TAD Z R		/R
2656	4020	JMS Z PRXLOP		
2657	1076	TAD Z A		/A
2660	4020	JMS Z PRXLOP		
2661	4323	JMS RSPACE		/R SP
2662	1055	TAD Z RITLNK		
2663	7440	SZA		
2664	5307	JMP LNONEX		/RIT LINK EQUALS A ONE
2665	4302	JMS ZEROR		
2666	1102	R01XX,	TAD Z SP	/SP
2667	4020	JMS Z PRXLOP		
2670	1054	TAD Z RARRTR		
2671	3057	DCA Z PROUT		
2672	5216	JMP COUNPR		/PRINT RARR TR CONTENTS
2673	2465	FINPRN,	FINPR	
2674	2507	R01XR,	R01X	
2675	2744	RTLPRX,	RTLPR	

2676	7240	LNONE,	CLA CMA	
2677	0106	AND Z ONE		
2700	4020	JMS Z PRXLOP		/PRINT LINK
2701	5674	JMP I R01XR		
2702	0000	ZEROR,	Ø	
2703	7240	CLA CMA		
2704	0105	AND Z ZERO		
2705	4020	JMS Z PRXLOP		/PRINT Ø LINK
2706	5702	JMP I ZEROR		
2707	7200	LNONEX,	CLA	
2710	1106	TAD Z ONE		
2711	4020	JMS Z PRXLOP		
2712	5266	JMP R01XX		
2713	0000	RTCRLF,	Ø	
2714	7200	CLA		
2715	4041	JMS Z CRLF		/CR LF
2716	1075	TAD Z R		/R
2717	4020	JMS Z PRXLOP		
2720	1100	TAD Z T		/T
2721	4020	JMS Z PRXLOP		
2722	5713	JMP I RTCRLF		
2723	0000	RSPACE,	Ø	
2724	7200	CLA		
2725	1075	TAD Z R		/R
2726	4020	JMS Z PRXLOP		
2727	1102	TAD Z SP		/SP
2730	4020	JMS Z PRXLOP		
2731	5723	JMP I RSPACE		

2732	7200	ROT2PR,	CLA	
2733	1275	TAD	RTLPRX	
2734	3200	DCA	PATPR	
2735	4313	JMS	RTCRLF	/CR LF RT
2736	4323	JMS	RSPACE	/R SP
2737	1055	TAD	Z RITLNK	
2740	7440	SZA		
2741	5307	JMP	LNONEX	/RIGHT LINK EQUALS A ONE
2742	4302	JMS	ZEROR	/PRINT 0 LINK
2743	5266	JMP	RO1XX	/PRINT SP AND RARRTR CONTENTS
2744	7200	RTLPR,	CLA	
2745	1273	TAD	FINPRN	
2746	3200	DCA	PATPR	
2747	4313	JMS	RTCRLF	/CR LF RT
2750	1077	TAD	Z L	/L
2751	4020	JMS	Z PRXLOP	
2752	1102	TAD	Z SP	/SP
2753	4020	JMS	Z PRXLOP	
2754	1053	TAD	Z LFTLNK	
2755	7440	SZA		
2756	5276	JMP	LNONE	/PRINT 1 LINK
2757	4302	JMS	ZEROR	/PRINT 0 LINK
2760	5674	JMP	I RO1XR	
2761	0000	PLINK,	0	
2762	1140	TAD	Z LNKSTR	/PRINT PAT LINK
2763	4237	JMS	ONZER	
2764	5761	JMP	I PLINK	

@A	0076	K	0110
ADD	0214	K0001	0073
ADDISM	0235	K0002	0072
ADDX	4050	K0004	0071
AIMS	0244	K0010	0070
ARGXXX	0640	K0020	0067
ARITH	2150	K0040	0066
ARITHT	0200	K0100	0065
AXISM	0265	K0200	0064
B	0114	K0400	0063
BISM	0272	K1000	0062
BITSTR	0125	K2000	0061
BW1	0117	K4000	0060
CISM	0305	K7377	0141
CL	0152	L	0077
CLLINK	2111	LCOMP	4077
CLRLNK	2015	LF	0103
CNTR	0123	LFTLNK	0053
CONHFD	0561	LINK	0134
CONTX	2414	LIST	4200
CONTX1	2445	LISTX	4015
CONTX2	2451	LNKSTR	0140
CONT1	2001	LNONF	2676
CONT2	2075	LNONFR	2516
COUNPR	2616	LNONFX	2707
COUNT	0111	LPXX	0022
COUNTX	0137	LRX	4122
COUNXX	2520	LRX	4123
CR	0104	LSTBIT	2621
CRLF	0041	MANYSP	0562
CRLFLF	0026	MESSG	0466
CRY	0120	MINDFX	2631
CX	0142	M144	4014
		NBIT	0472
		NOERR	0407
CXM	4075		
CXN	4074	NOERX	4124
D	0131	O	0107
DISM	0302	ODEVFN	4017
ERIN	2415	ONE	0106
ERROR	0400	ONEP	2644
ERRSW1	2400	ONZER	2637
ERRSW2	2406	P	0101
ERSW	2416	PADDOK	0312
ERSWIX	2115	PAT	0051
ERSW2X	2252	PATPR	2600
ERX	4076	PATPRX	2514
EX	2152	PLINK	2761
EX1	2163	PRERR	0423
FCOMP	4051	PRHEAD	0521
FINPR	2465	PRINT	0416
FINPRN	2673	PRONF	0506
FINPRX	2453	PROUT	0057
G	0130	PRSPAC	0512
GEN1	2000	PRXLOP	0020
GEN1X1	2446	PRZERO	0504
GEN2	2074	PTOTAL	0436
GEN2X2	2447	PT1EX	2011
HEADER	0124	PT1EXX	2105
HEDCON	0600	R	0075
HEDRJ	0563	RALRTL	0052
HSEKP	2116	RAND	4020
INCR	0223	RAND1	4030
INCRT	0415	RAND2	4000
INCRX	0224		

RARPR	2650
RARPRX	2454
RARRTR	0054
RCNT	4016
REROT1	2027
REROT2	2207
RITLTK	0055
RJHED	0625
ROPR	2455
ROTATE	0336
ROTK	2132
ROTX1	2444
ROTX2	2450
ROT1	2020
ROT1PR	2466
ROT2	2200
ROT2PR	2732
ROT2PX	2515
ROT2X	2114
RO1X	2507
RO1XR	2674
RO1XX	2666
RSPACE	2723
RTCRLF	2713
RTLPR	2744
RTLPRX	2675
SETLTK	2072
SL	0150
SP	0102
SPAC06	0126
SPACST	0127
SPA06	0530
SPA12	0632
STLTK	2250
STRCNT	0112
STRXY	0225
SUM	0122

SWOK	2434
SWOKX1	2521
SWOKX2	2525
SXOKX1	2151
SXOKX2	2253
SXY	0414
T	0100
TOTAL	0121
TST1	0056
TWELVE	0626
TWO	0113
TWORO	2464
TWOROX	2452
WD1	0115
WD2	0116
X	0132
XARG	0135
XFCOMP	0337
XONZFR	0435
XPROUT	0074
XSTRXY	4047
XTOTAL	0451
XZEROR	0564
Y	0133
YARG	0136
ZERO	0105
ZEROR	2702
ZERORX	2517
0	